Using Real-World Data

Realistic scenarios

Create DB using data from multiple sources.

Import data into existing DB.

Potential design issues

- Unclear keys, constraints, and relationships
- Use existing or new organization?

ETL

- Extract
- Transform
  - Organization
  - Data
- Load

Extract

- Conceptually straight-forward.
- Unknown file format?
- Lack compatible hardware?

Transform organization

- Get data appropriate source or table (join)
- Ignore some tables or attributes?
- Exchange rows & columns (pivot)
- Split or join columns
- Compute

Transform data

- Different units, formats, character sets, types
- Encoded
- Incorrect, missing, inconsistent, out of range

How to validate?
What to do if validation fails?

- What can be automated?

Load

- Conceptually straight-forward.
- Frequency?
- Auditing?